18/12/2020

Aakash
Medical | IIT-JEE | Foundations
(Divisions of Aakash Educational Services Limited)
***Regd. Office*:** Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph. 011-47623456

MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - I

Topics covered :

Physics **:** Physical World, Units & Measurements, Motion in a Straight Line, Motion in a Plane

Chemistry **:** Some Basic Concepts of Chemistry, Structure of Atom

Botany **:** The Living World, Biological Classification

Zoology **:** Animal Kingdom

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. Which among the given statements may be incorrect?

   (1) Law of conservation of energy is valid across all domains of nature.

   (2) The electromagnetic and weak nuclear forces have been unified, and are seen as aspects of single electroweak force

   (3) The strong nuclear force is strongest of all fundamental forces

   (4) Symmetry of nature with respect to translation in time is equivalent to law of conservation of linear momentum

2. In two columns C_1 and C_2 below which one is mismatched.

	C_1 Technology		C_2 Scientific principle
(1)	Aeroplane	–	Archimede's principle
(2)	Electron microscope	–	Wave nature of electron
(3)	Fusion test reactor	–	Magnetic confinement of plasma
(4)	Production of ultra high magnetic fields	–	Superconductivity

3. The angular diameter of near planet is found to be 2000″ from earth. The distance of the planet from earth is 1.6×10^{11} m. Then diameter of the planet is

   (1) 1.55×10^{10} m (2) 1.55×10^{9} m

   (3) 2.4×10^{9} m (4) 1.95×10^{8} m

4. True length of a rod is 3.652 cm. In one experiment, the measured value is found to be 3.6 cm while in another experiment the length is determined as 3.48 cm then
   (1) First measurement is more accurate and more precise
   (2) Second measurement is more accurate and more precise
   (3) First measurement is more accurate but less precise
   (4) Second measurement is less accurate and less precise

5. Which among the following statements may be **incorrect**?
   (1) Systematic errors can be minimised by improving experimental techniques
   (2) Error arising due to imperfect calibration of instrument is a systematic error
   (3) The random errors are those errors which occur irregularly.
   (4) Least count errors always belong to category of random errors

6. The time period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of length is 40.0 cm known to 2 mm accuracy and time for 100 oscillation of a pendulum is found to be 80 s using a wrist watch of 1 second resolution. The accuracy in determination of g is
   (1) 2% (2) 2.5%
   (3) 3% (4) 4.5%

7. 9.85 g of substance occupies volume of 1.2 cm^3. The density of the substance is (with significant figures in view)
   (1) 8.2 g cm^{-3} (2) 8.21 g cm^{-3}
   (3) 8.20 g cm^{-3} (4) 8.208 g cm^{-3}

8. A physical quantity Q is related to three observable quantity x, y, z as follows $Q = \frac{x\sqrt{y}}{z^2}$. The percentage error in measurement of x, y, z are 5%, 4% and 1% respectively. The maximum percentage error in the quantity Q is
   (1) 9% (2) 7%
   (3) 5% (4) 3%

9. Van der Waal's equation is $\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$, where P, V, R, T and n represents the pressure, volume, universal gas constant, absolute temperature and number of moles of the gas respectively. Then dimensions of $\left[\frac{a}{b^2}\right]$ is of
   (1) Force (2) Momentum
   (3) Energy (4) Pressure

10. Which of the following pairs have same dimensional formula for both the quantities?
   (A) Resistance and inductance
   (B) Kinetic energy and torque
   (C) Planck's constant and momentum
   (D) Pressure and Bulk modulus
   (1) (B) only
   (2) (C) only
   (3) (B) and (D) only
   (4) (A), (B) and (D) only

11. A person walks along a circular path of radius 150 m at a constant pace. He complete one circle in 10 minutes. In any 5 minutes of duration, his average velocity is
   (1) 2 m s^{-1} (2) 2π m s^{-1}
   (3) $\frac{\pi}{2}$ m s^{-1} (4) 1 m s^{-1}

12. A body is projected upwards with a velocity of 98 m/s. Its speed at a height of 196 m from the point of projection is nearly (g = 9.8 m s^{-2})
   (1) 40 m s^{-1}
   (2) 76 m s^{-1}
   (3) 96 m s^{-1}
   (4) Zero

13. A particle starting from rest has a velocity that increases linearly with time $v = \frac{t}{2}$, where v is in m s^{-1} and t in s. The distance travelled in first four seconds is
   (1) 9 m
   (2) 4 m
   (3) 8 m
   (4) 32 m

14. From a 100 m high tower, a ball is thrown upwards with a speed of 15 m s^{-1} and another is thrown vertically downward at same speed simultaneously. The time difference of their reaching the ground will be (g = 10 m s^{-2})

(1) 2 s (2) 4 s

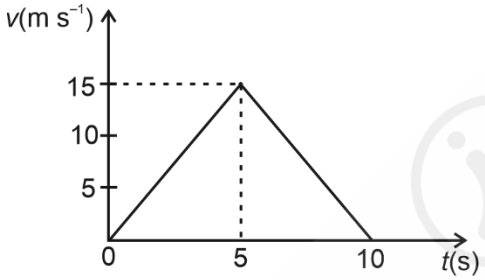
(3) 1 s (4) 3 s

15. A train of length 300 m travelling at 30 m s^{-1} crosses another train of length 200 m travelling at 20 m s^{-1} in opposite direction. Time taken by first train to pass the second train is

(1) 10 s (2) 30 s

(3) 50 s (4) 2 s

16. Speed-time graph of a particle moving along a fixed direction is as shown in figure. The distance travelled by particle in first 7 s is



(1) 40 m (2) 75 m

(3) 61.5 m (4) 55.5 m

17. The position of a particle with respect to time along x-axis is given by $x = 6t^2 - t^3$, where x in metre and t in s. What will be position of particle when it achieves maximum velocity?

(1) 24 m (2) 32 m

(3) 9 m (4) 16 m

18. A car moving along a straight track with speed of 108 km/h is brought to stop within a distance of 105 m. How long does it take for the car to stop under constant retardation?

(1) 7 s (2) 9 s

(3) 10 s (4) 12 s

19. The position of an object moving along x-axis is given by x = 8.5 + 2.5 t^2, x in metre and t in second. The average velocity of object between t = 2 s and t = 4 s is

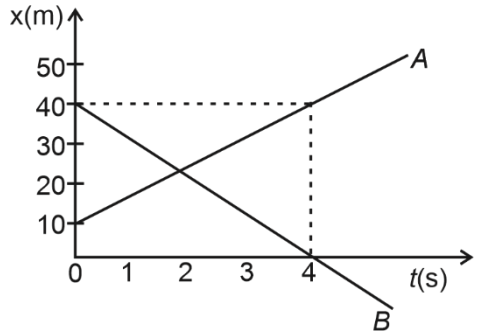
(1) 10 m s^{-1}

(2) 12 m s^{-1}

(3) 15 m s^{-1}

(4) 18 m s^{-1}

20. The position-time graph for two objects during motion is shown in diagram. The magnitude of relative velocity of A w.r.t. B is



(1) 7.5 m s^{-1}

(2) 17.5 m s^{-1}

(3) 2.5 m s^{-1}

(4) 20 m s^{-1}

21. Rain is falling vertically with a speed of 35 m s^{-1}. Wind starts blowing after sometimes from south to north with a speed of 12 m s^{-1}. The speed with which rain drops hit a boy waiting at open bus stop, is

(1) 47 m s^{-1}

(2) 37 m s^{-1}

(3) 42 m s^{-1}

(4) 43.7 m s^{-1}

22. The position of a particle in two dimensional motion is given as $\vec{r} = 4t\hat{i} + 4t^2\hat{j}$, where t is in second and r in metre, then direction of motion of particle at t = 1 second with x-axis is

(1) $\theta = \tan^{-1}(4)$

(2) $\theta = \tan^{-1}(3)$

(3) $\theta = \tan^{-1}(2)$

(4) $\theta = \tan^{-1}(1/2)$

23. A particle starts from origin at t = 0 with velocity of $2\hat{i}$ m s^{-1} and moves in x-y plane under a constant acceleration of $(3\hat{i} + 2\hat{j})$ m s^{-2}. What is x co-ordinate of the particle where its y co-ordinate is 16 m?

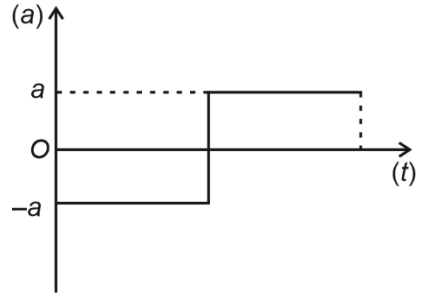
(1) 28 m

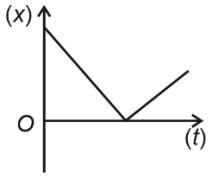
(2) 32 m

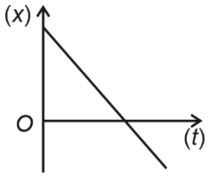
(3) 16 m

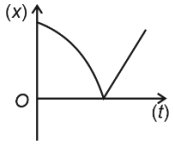
(4) 48 m

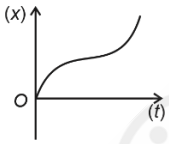
24. The acceleration-time graph for an object in motion along a straight line is shown in figure. The corresponding position-time graph may be represented by



(1) 

(2) 

(3) 

(4) 

25. Stopping distance of moving vehicle is directly proportional to

(1) Square of its acceleration

(2) Square of its initial velocity

(3) Initial velocity

(4) Initial velocity and acceleration

26. A ball *A* is dropped from a building of height 78.4 m, simultaneously another identical ball *B* is thrown up with speed of 49.0 m s^{-1}. The relative velocity of ball *B* w.r.t. ball *A*, 3 second later is (g = 9.8 m s^{-2})

(1) 19.6 m s^{-1} (2) 29.4 m s^{-1}

(3) 49.0 m s^{-1} (4) Zero

27. At a metro mall, a girl walks up a stationary escalator in 120 second. If she remains stationary on the escalator then the escalator take her up in 4 minute. The time taken by her to walk up on the moving escalator will be

(1) 140 s (2) 200 s

(3) 100 s (4) 80 s

28. Two vectors $\vec{A}$ and $\vec{B}$ are inclined at an angle θ have their resultant $\vec{R}$ which makes an angle α with $\vec{A}$. If the direction of $\vec{A}$ and $\vec{B}$ are interchanged then the resultant will have

(1) Same direction

(2) Same magnitude

(3) Different magnitude

(4) Same magnitude and direction

29. The (x, y, z) co-ordinates of two points *A* and *B* are given respectively as (0, 4, –2) and (–2, 8, –4). The displacement vector from *B* to *A* is

(1) $2\hat{i} - 4\hat{j} + \hat{k}$

(2) $-2\hat{i} + 4\hat{j} - 2\hat{k}$

(3) $2\hat{i} - 4\hat{j} + 2\hat{k}$

(4) $2\hat{i} - 4\hat{j} - 2\hat{k}$

30. A river is flowing from west to east with speed of 4 m s^{-1}. A swimmer can swim in still water at a speed of 4 m s^{-1}. If he wants to starts from a point *P* on south bank and reach exactly a point *Q* opposite on north bank, in what direction should he swim?

(1) 30° east to north

(2) 45° east to north

(3) Towards north

(4) He will never reach point *Q* by swimming

31. A stone is thrown horizontally from a cliff of height 490 m above ground with initial speed of 15 m s^{-1}. The speed with which stone will hit the ground, is nearly (g = 9.8 m s^{-2})

(1) 98 m s^{-1} (2) 99 m s^{-1}

(3) 110 m s^{-1} (4) 15 m s^{-1}

32. A football is kicked at a speed of 28 m s^{-1} in a direction 30° above the horizontal. Its speed 1.4 second later is (g = 10 m s^{-2})

(1) 24.2 m s^{-1}

(2) 16.4 m s^{-1}

(3) 9.6 m s^{-1}

(4) 22.6 m s^{-1}

33. An insect is moving on a circular path of radius 25 cm. It moves steadily and completes 7 revolutions in 44 seconds. What is magnitude of acceleration of its motion?

(1) 1 m s^{-2}

(2) 0.75 m s^{-2}

(3) 1.5 m s^{-2}

(4) 0.25 m s^{-2}

34. A gun shot is fired horizontally above ground with initial speed of 300 m s^{-1}. When it falls through a vertical distance of 5 cm, the horizontal distance covered is (g = 10 m s^{-2})

(1) 15 m (2) 30 m

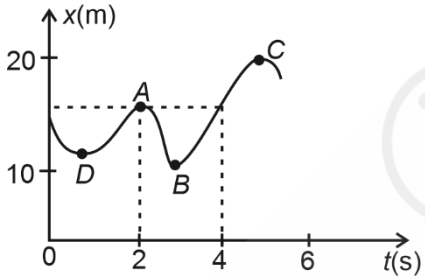
(3) 32 m (4) 18 m

35. A pebble is released into a well of depth h. The splash is heard after time t second of release. If v is velocity of sound in air, then

(1) $t = \frac{gh^2}{v}$ (2) $t = \sqrt{\frac{2h}{g}} + \frac{h}{v}$

(3) $\sqrt{\frac{h}{g}} + \frac{2h}{v}$ (4) $\frac{2h}{v}$

36. Figure shows the position of particle moving along x-axis as a function of time. Which of the following statement is correct?



(1) The particle has come to rest 3 times

(2) The maximum speed is at C

(3) The average velocity for total period shown is positive

(4) Velocity remains positive for first 4 second of motion

37. A point from where a ball is projected is taken as origin of co-ordinate axis. x and y component of its displacement are given by $x = 6t$ and $y = 10t - 5t^2$, where x and y are in meter and t in second. What is the range of this projectile?

(1) 12 m

(2) 18 m

(3) 60 m

(4) 30 m

38. What is angle of projection of a body with the horizontal when its horizontal range is $\frac{4}{3}$ times the maximum height reached?

(1) $\tan^{-1}(2)$ (2) $\tan^{-1}(3)$

(3) $\tan^{-1}(1/3)$ (4) $\tan^{-1}(1/2)$

39. A ball is thrown from a point on ground at different angles with same speed of 30 m s^{-1} and has the same range in two cases. If h_1 and h_2 are maximum heights attained in these cases then ($h_1 + h_2$) is equal to (g = 10 m s^{-2})

(1) 45 m

(2) 90 m

(3) 22.5 m

(4) 60 m

40. A cyclist is riding with speed of 27 km h^{-1}. As he approaches a circular turn on the road of radius 90 m, he applies brakes and reduces his speed at a constant rate of 1 m s^{-2}. What is the magnitude of acceleration of the cyclist on just entering circular turn?

(1) 0.375 m s^{-2}

(2) 1.625 m s^{-2}

(3) 1.18 m s^{-2}

(4) 1.72 m s^{-2}

41. If the time of flight of a projectile is doubled, the maximum height attained will be (initial speed remains same)

(1) Doubled

(2) Four times

(3) One fourth

(4) Remains same

42. What is magnitude of resultant of three coplanar forces F_1 = 300 N at 0°, F_2 = 400 N at 30°, F_3 = 400 N at 150° with x-axis?

(1) 700 N

(2) 500 N

(3) 1100 N

(4) 300 N

43. Taking horizontal direction as x-axis and vertical upwards as positive y-axis. A ball is projected from origin with initial velocity of $\vec{u} = 20\,\hat{i} + 10\,\hat{j}$ m s^{-1}. The magnitude of change of velocity of ball till it reaches back the level of projection is (g = 10 m s^{-2})

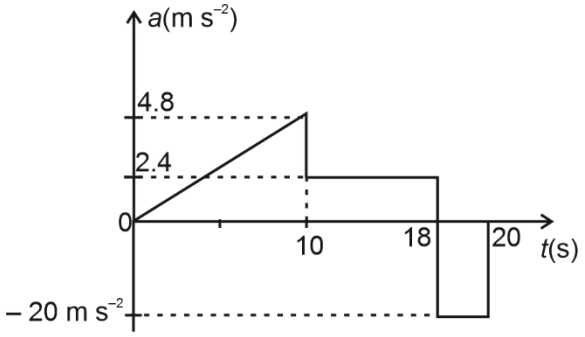
(1) 40 m s^{-1}

(2) 20 m s^{-1}

(3) 60 m s^{-1}

(4) 0 m s^{-1}

44. Acceleration-time graph for an object during motion is shown below.



In which time duration, magnitude of change of velocity will be maximum?

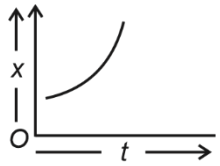
(1) t = 0 till t = 10 s

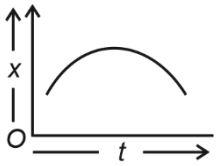
(2) t = 10 s till t = 18 s

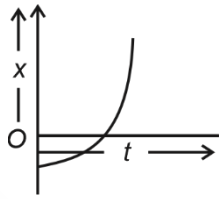
(3) t = 18 s till t = 20 s

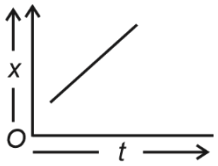
(4) More information is required

45. Position-time graph for four objects *A*, *B*, *C*, and *D* during their motion are shown in figures below. In which case the object has constant negative acceleration throughout its motion.

A. 

B. 

C. 

D. 

(1) A (2) C

(3) D (4) B

CHEMISTRY

46. Number of atoms present in 0.2 g molecule of H_2SO_4 is (N_A represents Avogadro's number)

(1) 1.4 (2) 1.4 N_A

(3) 2.8 (4) 2.8 N_A

47. Molarity of the solution in which 4 g NaOH is present in 250 ml solution, is

(1) 0.4 M (2) 0.2 M

(3) 0.3 M (4) 0.8 M

48. Volume of water added to convert 0.5 L aqueous solution of 0.3 M glucose into 0.2 M glucose, is

(1) 0.75 L (2) 0.5 L

(3) 0.25 L (4) 0.4 L

49. In an aqueous solution, mole fraction of $C_6H_{12}O_6$ is 0.2. Molality of this solution is

(1) 11.33 m

(2) 13.89 m

(3) 26.55 m

(4) 1.75 m

50. If phosphoric acid (H_3PO_4) on reaction with NaOH forms Na_2HPO_4; then equivalent mass of phosphoric acid in the reaction will be

(1) 50 (2) 49

(3) 98 (4) 196

51. What is the mass of the precipitate formed when 50 ml of 20.8% (w/v) solution of $BaCl_2$ is mixed with 50 ml of 14.2 % (w/v) Na_2SO_4 solution? (g atomic mass of barium = 137 g mol^{-1})

(1) 10 g (2) 11.65 g

(3) 23.3 g (4) 5.825 g

52. Suppose the elements X and Y combine to form two compounds XY and X_2Y_3. When 0.1 mol of XY weighs 10 g and 0.05 mole of X_2Y_3 weighs 12 g, then the atomic weights of X and Y respectively are

(1) 50, 50 (2) 20, 80

(3) 60, 40 (4) 30, 70

53. 4 g of calcium is burnt with 0.64 g of oxygen gas in a closed vessel. Which reactant is left in excess and how much?

(1) Ca, 1.2 g (2) Ca, 2.4 g

(3) Ca, 3.2 g (4) O_2, 0.32 g

54. Volume occupied by one molecule of a liquid (density of liquid = 1.2 g cm^{-3} and molar mass of liquid is 60 g mol^{-1}) will be

(1) 3×10^{-24} cm^3

(2) 3.33×10^{-24} cm^3

(3) 8.33×10^{-23} cm^3

(4) 8.33×10^{-29} cm^3

55. What volume of air measured at 0°C and 1 atm, is needed to burn completely 1 L of butane gas (C_4H_{10}) measured under the same conditions? (Assume air has 20% oxygen)

(1) 16.5 L (2) 3.25 L

(3) 32.5 L (4) 1.65 L

56. If percentage abundance of three isotopes of an atom A of atomic masses 20, 18 and 22 respectively are 95%, 4% and 1%, then the weighted average atomic mass of the naturally occurring element A will be

(1) 15 amu (2) 16.5 amu

(3) 19.94 amu (4) 21.34 amu

57. Concentrated aqueous sulphuric acid is 73.5% H_2SO_4 by mass and has a density of 1.2 g ml^{-1}. Volume of this acid solution required to form 0.81 L of 2 M H_2SO_4 is

(1) 120 ml (2) 150 ml

(3) 180 ml (4) 90 ml

58. The total number of valence electrons in 4.8 g of O^{2-} ion is (N_A is Avogadro's number)

(1) 0.2 N_A (2) 0.3 N_A

(3) 2 N_A (4) 2.4 N_A

59. Change in volume when 200 ml HI decomposed on gold surface into $H_2(g)$ and $I_2(g)$ is

(1) Increase by 20 ml

(2) Increase by 100 ml

(3) Zero

(4) Decreased by 50 ml

60. In vulcanised rubber, sulphur is present by 0.016% by weight, then minimum molecular weight of vulcanised rubber will be

(1) 2×10^4 (2) 2×10^5

(3) 1×10^4 (4) 4×10^5

61. Mole fraction of urea in aqueous solution of 10% by mass of urea, is

(1) 0.079 (2) 0.032

(3) 0.095 (4) 0.735

62. Number of significant figures in the addition of 3.24, 4.263 and 4.3 is

(1) 1 (2) 2

(3) 3 (4) 4

63. If 1.5 L of 0.4 M NaOH solution is mixed with another 2 L of 0.3 M NaOH solution, then final molarity of resultant solution is

(1) 0.02 M (2) 0.34 M

(3) 0.04 M (4) 0.93 M

64. From 230 mg of N_2O_4, x molecules are removed, 5×10^{-4} mole of N_2O_4 is left, then x will be

(1) 2×10^{-3} (2) 1.2×10^{21}

(3) 4×10^{-3} (4) 2.4×10^{21}

65. If x g of metal sulphate contains y g of metal, then equivalent mass of metal will be

(1) $\frac{48\,x}{y}$ (2) $\frac{48\,y}{(x+y)}$

(3) $\frac{48\,x}{x-y}$ (4) $\frac{48\,y}{x-y}$

66. Concentration term which depends on temperature, is

(1) Molality (2) Mole fraction

(3) % w/v (4) % w/w

67. In a compound, mass of carbon present is four times of hydrogen while mass of oxygen present is 1.5 times of carbon. Empirical formula of the compound will be

(1) C_4H_2O

(2) $C_8H_{16}O$

(3) $C_8H_{24}O_9$

(4) $C_8H_{30}O_2$

68. 30 g of a calcium carbonate sample decomposes on heating to give 12 g calcium oxide. What will be the percentage purity of calcium carbonate in the sample? (At. Wt. Ca = 40)

(1) 40% (2) 71.4%

(3) 52.2% (4) 24.2%

69. Which set of orbitals will have electron density along the axes?

(1) p_x, p_y (2) d_{xy}, d_{yz}

(3) d_{yz}, d_{zx} (4) $d_{x^2-y^2}$, d_{xy}

70. Maximum number of electrons that can fit in the orbital(s) for which n = 4 and ℓ = 1?

(1) 4 (2) 10

(3) 6 (4) 14

71. Among the following the orbital angular moment of which orbital will be highest?

(1) 2 s (2) 3 p

(3) 5 p (4) 3 d

72. If the uncertainity in position is 4 times of uncertainity in momentum then uncertainity in velocity is

(1) $\frac{1}{2}\sqrt{\frac{h}{\pi}}$ (2) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

(3) $\frac{1}{4m}\sqrt{\frac{h}{\pi}}$ (4) $\frac{1}{4m}\sqrt{\frac{h^2}{\pi}}$

73. Correct set of quantum numbers for last electron in potassium atom is

(1) n = 4, ℓ = 1, m = 0, s = +1/2

(2) n = 4, ℓ = 0, m = 0, s = +1/2

(3) n = 3, ℓ = 1, m = 0, s = +1/2

(4) n = 4, ℓ = 0, m = –1, s = +1/2

74. Among the following which transition in the hydrogen atom will give rise to the highest energetic photon?

(1) n = 6 to n = 5 (2) n = 5 to n = 4

(3) n = 4 to n = 3 (4) n = 3 to n = 1

75. de-Broglie wavelength associated with a microscopic particle of 10^{-20} mg mass and having speed of 50 m/s, is (h = 6×10^{-34} J-s)

(1) 12 Å (2) 120 Å

(3) 20 Å (4) 200 Å

76. The radius ratio of 3rd orbit of Li^{2+} ion to 4th orbit of Be^{3+} ion is

(1) 1 : 1 (2) 2 : 1

(3) 3 : 4 (4) 1 : 3

77. Number of spectral lines of Balmer series obtained when electrons are de-excited from n^{th} shell to ground state in sample of H-atoms is

(1) n – 1 (2) n – 2

(3) n – 3 (4) n – 4

78. Which one of the following ion has electronic configuration of [Ar] $3d^8$?

(1) Cr^{3+} (2) Fe^{2+}

(3) Co^{3+} (4) Ni^{2+}

79. If the energy of second Bohr orbital of Li^{2+} ion is –x kJ/mol then energy of third Bohr orbit of Be^{3+} ion (in kJ/mol) will be

(1) –64 x (2) $-\frac{64}{9}x$

(3) –2x (4) $-\frac{64}{81}x$

80. Select the orbital which will be filled first (in multielectron species)

(1) 6s (2) 4f

(3) 3d (4) 5p

81. Which of the following species is isoelectronic with CO?

(1) NO (2) CN^-

(3) O_2^+ (4) N_2^+

82. If the energy given by photons to a metal is 40% higher than its threshold energy then maximum kinetic energy of photoelectrons will be

(1) 4 times of work function

(2) 0.286 times of energy of photons

(3) 0.5 times of energy of photons

(4) 0.75 times of work function

83. An ion M^{x+} (atomic number of 27) has magnetic moment equal to 3.87 BM. The value of x is equal to

(1) 2 (2) 3

(3) 4 (4) 6

84. If radius of first orbit of hydrogen atom is x Å then wavelength associated with electron in 3rd orbit of He^+ ion is

(1) 2πx (2) 4πx

(3) 3πx (4) 6πx

85. Series limit wavelength of Paschen series in H-atom is

(1) 9R (2) R/9

(3) 9/R (4) 81R

86. Correct order of energy of 2s orbital in the following monovalent cationic species is

(1) E_{2s} (Be^+) > E_{2s} (Mg^+) > E_{2s} (Ca^+) > E_{2s} (Sr^+)

(2) E_{2s} (Be^+) > E_{2s} (Sr^+) > E_{2s} (Ca^+) > E_{2s} (Mg^+)

(3) E_{2s} (Sr^+) > E_{2s} (Ca^+) > E_{2s} (Mg^+) > E_{2s} (Be^+)

(4) E_{2s} (Sr^+) > E_{2s} (Be^+) > E_{2s} (Mg^+) > E_{2s} (Ca^+)

87. Number of radial and angular nodes associated with $3d_{xy}$ orbital respectively are
(1) 2, 2
(2) 0, 2
(3) 3, 2
(4) 2, 1

88. Wave number (in cm^{-1}) associated with yellow radiation having wavelength of 6000 Å is
(1) 1.67×10^6
(2) 1.67×10^4
(3) 2.82×10^4
(4) 2.82×10^6

89. Correct order of frequency of electromagnetic radiations is
(1) γ-rays > X-rays > UV > IR > Microwave
(2) γ-rays > UV > IR > Microwave > X-rays
(3) X-rays > UV > IR > Microwave > γ-rays
(4) X-rays > IR > Microwave > γ -rays > UV

90. Select the correct statement among the following
(1) *s*-orbital has one nodal plane
(2) *p*-orbital has one nodal plane
(3) *d*-orbitals have dumbbell shape
(4) *f*-orbitals are spherically symmetric

BOTANY

91. Defining feature exclusive to human beings is
(1) Metabolism
(2) Cellular organisation
(3) Self consciousness
(4) Intrinsic growth

92. A binomial epithet of a species includes all, **except**
(1) Genus name
(2) Specific epithet
(3) Author name in abbreviated form
(4) Sub species name

93. The common taxonomic category shared by both wheat and brinjal is
(1) Class
(2) Order
(3) Division
(4) Family

94. Select the **incorrect** match w.r.t. classification of man.
(1) Family – Hominidae
(2) Phylum – Mammalia
(3) Order – Primata
(4) Genus – *Homo*

95. Scientific names of prokaryotes are standardized by
(1) ICBN (2) ICNCP
(3) ICNB (4) ICVCN

96. According to binomial nomenclature, scientific names
(a) Are generally taken from Latin language
(b) Consist of two words
(c) Have first word denoting specific epithet
The **correct** one(s) is/are
(1) (a) only (2) (b) only
(3) (a) and (b) (4) (b) and (c)

97. Three genera *Datura*, *Petunia* and *Solanum* share common vegetative and reproductive characteristics. On this basis they are placed in same
(1) Order (2) Family
(3) Class (4) Division

98. The taxonomic aid called store house of preserved plant specimens is
(1) Flora
(2) Herbarium
(3) Botanical garden
(4) Manual

99. Keys are
(1) Used for identification of plants only
(2) Based on contrasting characters
(3) Collections of specimens of organisms
(4) Useful to have complete description along with detail classification of an organism

100. _____ provide index to the plant species found in a particular area.

Choose the **correct** option to fill up the above blank.

(1) Manual (2) Monograph

(3) Flora (4) Museum

101. According to five kingdom classification system, how many kingdoms do /does **not** include(s) any autotrophic organism?

(1) 3 (2) 4

(3) 1 (4) 2

102. Mark these statements as true **(T)** or false **(F)** and select the **correct** option.

A. In three domain system, all eukaryotes are divided into two domains.

B. Comma-shaped bacteria are called vibrio.

C. Kingdom Monera shows the most extensive metabolic diversity.

(1) A(T), B(T), C(T) (2) A(T), B(T), C(F)

(3) A(F), B(T), C(T) (4) A(F), B(F), C(T)

103. Which of the given statements is **not** true for methanogens?

(1) Responsible for production of biogas

(2) Have different cell wall structure than that of other bacteria

(3) Found in gut of several ruminants

(4) These are considered as true bacteria

104. "Chlorophyll a appeared first in the group called A and pigments identical to higher plants are found in B .

Complete above statement by choosing **correct** option for A and B.

(1) A-Diatoms, B-Cyanobacteria

(2) A-Purple bacteria, B-BGA

(3) A-Cyanobacteria, B-Euglenoids

(4) A-Euglenoids, B-Golden algae

105. Chief producers in oceans

(1) Have soap box like body

(2) Are found in marine environment only

(3) Lack cell wall

(4) Have flagella in their body

106. Generally *Nostoc* lacks

(1) Heterocysts

(2) Gelatinous sheath

(3) Photosynthetic pigments

(4) Heterotrophic mode of nutrition

107. All of the following are due to activities of heterotrophic bacteria, **except**

(1) Conversion of milk to curd

(2) Fixing nitrogen in pea

(3) Antibiotic production

(4) Release of oxygen during metabolism

108. Wall-less moneran is

(1) *Anabaena* (2) *Mycoplasma*

(3) Halophiles (4) *E. coli*

109. Flagella are absent in

(1) *Gonyaulax* (2) *Euglena*

(3) *Trypanosoma* (4) *Entamoeba*

110. *Paramoecium* does **not** have

(1) Gullet for taking up food

(2) Two types of nuclei

(3) Cilia on surface of cell

(4) Silica shell on its surface

111. A. Slime moulds form fruiting bodies under unfavourable conditions.

B. Euglenoids may predate on smaller organisms.

Read above statements and choose the option accordingly.

(1) Only A is correct

(2) Only B is correct

(3) Both A and B are correct

(4) Both A and B are incorrect

112. Fungus which causes white rust disease on mustard leaves.

(a) Is a sac fungus

(b) Has coenocytic mycelium

(c) Lacks dikaryophase in its life cycle

The **correct** ones are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) All (a), (b), (c)

113. The fungus called bread mould is
(1) *Mucor* (2) *Rhizopus*
(3) *Penicillium* (4) Yeast

114. Match column I with column II and select the **correct** option.

	Column-I		**Column-II**
a.	*Puccinia*	(i)	Parasitic phycomycetes
b.	*Colletotrichum*	(ii)	Club fungus
c.	*Albugo*	(iii)	Smut fungus
d.	*Ustilago*	(iv)	Imperfect fungus

(1) a(ii), b(iv), c(iii), d(i)
(2) a(iv), b(i), c(iii), d(ii)
(3) a(iv), b(ii), c(iii), d(i)
(4) a(ii), b(iv), c(i), d(iii)

115. Edible members of Ascomycetes group are
(1) *Agaricus* and yeast
(2) Morels and truffles
(3) Yeast and *Ustilago*
(4) *Trichoderma* and truffles

116. Choose the **odd** one w.r.t. Basidiomycetes.
(1) Puffball
(2) Bracket fungi
(3) *Puccinia*
(4) *Claviceps*

117. All of the following are means of vegetative reproduction in fungi, **except**
(1) Fission (2) Budding
(3) Fragmentation (4) Regeneration

118. Which of the given fungus is used extensively in biochemical and genetic work?
(1) *Neurospora* (2) *Alternaria*
(3) *Puccinia* (4) *Penicillium*

119. Toadstools are
(1) Known as common mushrooms
(2) Edible mushrooms
(3) Members of Basidiomycetes
(4) Unicellular

120. Select the **wrong** match.
(1) Zoospores – Asexual spores
(2) Conidia – Aplanospores
(3) Sporangiospores – Endogenously produced
(4) Ascospores – Exogenously produced asexual spores

121. Members of Basidiomycetes lack
(1) Sex organs
(2) Plasmogamy and karyogamy
(3) Fruiting bodies
(4) Meiospores

122. Read the following features.
(a) Help in mineral cycling.
(b) Lack karyogamy and meiosis
(c) Conidia are asexual spores.
Above given features are **true** for the member of group
(1) Ascomycetes
(2) Basidiomycetes
(3) Phycomycetes
(4) Deuteromycetes

123. *Agaricus* has/possesses
(1) Edible fruiting bodies
(2) Aseptate mycelium
(3) Motile asexual spores
(4) Motile meiospores

124. Members of kingdom plantae
(1) May include some prokaryotes
(2) Can be partial heterotroph
(3) Lack cellulosic cell wall
(4) Always have equal lengths of diploid and haploid phases in their life

125. Who called the extract of the infected plant of tobacco as *Contagium vivum fluidum*?
(1) Stanley
(2) Ivanowsky
(3) Beijerinck
(4) Pasteur

126. Bacterial viruses
(a) Have polyhedral head.
(b) Usually possess dsDNA.
(c) Have 6 tail fibres.
The **correct** ones are
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) All (a), (b) and (c)

127. Choose the **incorrect** one for viroids.
(1) Discovered by Diener
(2) Smaller than viruses
(3) Have free DNA or RNA
(4) Can cause diseases in plants

128. Mutually beneficial relation between algae and fungi is
(1) Known as lichen
(2) Easily found in air polluted area
(3) Considered as a single organism
(4) Both (1) and (2)

129. Prions are causal organisms of
(1) Mad cow disease
(2) Potato spindle tuber disease
(3) Influenza
(4) Tobacco mosaic disease

130. Prions are
(a) Similar in size to viruses
(b) Abnormally folded proteins
(c) Infectious to humans only
The **correct** one(s) is/are
(1) (a) and (b) only (2) (b) only
(3) (b) and (c) only (4) All (a), (b) and (c)

131. Who showed that viruses could be crystallised?
(1) Ivanowsky
(2) Leeuwenhoeck
(3) Stanley
(4) Randle

132. Choose the **odd** one w.r.t. viral diseases.
(1) Herpes
(2) Mumps
(3) Small pox
(4) Creutzfeldt-Jacob disease

133. A parasitic protozoan that has flagella is
(1) *Gonyaulax*
(2) A malaria causing organism
(3) *Amoeba*
(4) A sleeping sickness causing organism

134. Under unfavourable conditions bacteria reproduce by means of
(1) Binary fission
(2) Spores
(3) Zoospores
(4) Sexual recombination

135. Linnaeus kept all the organisms algae, bacteria, fungi, bryophytes, angiosperms together in single kingdom Plantae because of the presence of
(1) Eukaryotic cell
(2) Chlorophyll
(3) Cell wall
(4) Locomotory structure

ZOOLOGY

136. One of the representative of phylum Echinodermata is
(1) *Echinus* (2) *Ascidia*
(3) *Balanoglossus* (4) *Pila*

137. Which group of animals belong to the same phylum?
(1) Lancelet, Hagfish, Vulture
(2) Dog, Cat, Starfish
(3) Apple snail, Cuttlefish, King carb
(4) Earthworm, Hookworm, Roundworm

138. Metameric segmentation and true coelom are characteristic of phylum
(1) Annelida and Arthropoda
(2) Mollusca and Chordata
(3) Aschelminthes and Annelida
(4) Platyhelminthes and Arthropoda

139. Reproduction only by sexual means is characteristic of
(1) *Physalia* (2) *Adamsia*
(3) *Pleurobrachia* (4) *Euspongia*

140. What is **true** about *Betta*, *Myxine* and *Chelone*?
(1) They all possess dorsal heart
(2) They all exhibit internal fertilisation and direct development
(3) They all possess paired fins/limbs
(4) They all lack the capacity to regulate their body temperature

141. Identify the given diagram and select the statement which is **correct** for it.



(1) It possess notochord from head to tail in both larval and adult stage
(2) It has a pair of kidneys for excretion and osmoregulation
(3) It exhibits bilateral symmetry and metamerism
(4) It belongs to phylum Chordata and super class Tetrapoda

142. Which of the following characteristic is shared by Penguin and Turtle?
(1) Oviparity and indirect development
(2) Homoiothermy
(3) Pneumatic bones
(4) Cranium and vertebral column is bony

143. Which of the following animal is **correctly** matched with its zoological name and taxonomic category?

(1)	Angelfish	–	*Pterophyllum*, a class
(2)	Lamprey	–	*Petromyzon*, a division
(3)	Sawfish	–	*Pristis*, a genus
(4)	Tree frog	–	*Hyla*, a species

144. Match the name of the animal (Column-I), with one characteristic (Column-II), and the phylum/class (Column-III) to which it belongs

	Column-I	**Column-II**	**Column-III**
(1)	*Corvus*	Poikilotherms	*Aves*
(2)	*Vipera*	External ear openings are present	*Reptilia*
(3)	*Panthera*	Notochord is persistent throughout life	*Mammalia*
(4)	*Bufo*	External fertilisation	*Chordata*

145. Sexes are separate and fertilisation is internal in all of the following organisms **except**
(1) *Hippocampus*
(2) *Scoliodon*
(3) *Psittacula*
(4) *Naja*

146. Choose the **mismatch**.

(1)	*Labeo*	–	Sub-phylum Chordata
(2)	*Salamandra*	–	Super class Pisces
(3)	*Testudo*	–	Division Agnatha
(4)	*Macaca*	–	Sub-phylum Vertebrata

147. Mantle cavity contains a structure ctenidia, which is helpful in respiration. It is a characteristic of
(1) *Aedes* (2) *Echinus*
(3) *Loligo* (4) *Salpa*

148. Which of the following characteristic features hold **true** for the corresponding group of animals?

(1)	A small group of worm-like marine animals with organ-system level of organisation	Arthropoda

(2)	The adults are radially symmetrical but larvae are bilaterally symmetrical	Hemichordata
(3)	Notochord extends from head to tail region and is persistent throughout their life	Cephalochordata
(4)	Dorsal muscular heart with, two, three or four chambers	Vertebrata

149. Which characteristic is **not** shared by hemichordates and chordates?
(1) True coelom
(2) Bilateral symmetry
(3) Organ system level of organisation
(4) True segmentation

150. Which of the following characteristic is **not** applicable to *Antedon*?
(1) Complete digestive system with mouth on dorsal side and anus of ventral side
(2) Indirect development with free swimming larva
(3) Presence of water vascular system
(4) Endoskeleton of calcareous ossicles

151. How many animals given in the box belong to division Gnathostomata?

Doliolum, Branchiostoma, Clarias, Myxine, Macropus, Salpa, Saccoglossus

(1) Two
(2) Three
(3) Four
(4) Five

152. Mark the option in which all the mentioned animals show external fertilisation?
(1) *Balanoglossus, Labeo, Bufo*
(2) *Charcharodon, Scoliodon, Betta*
(3) *Ichthyophis, Myxine, Struthio*
(4) *Hemidactylus, Pristis, Echinus*

153. Choose the **correct** statement.
(1) Aquatic annelids like *Pila* possess lateral appendages, parapodia, which help in swimming
(2) The body of arthropods are covered by calcareous shell
(3) In sponges, the body is supported by skeleton made up of spicules or spongin fibres
(4) *Hydra* exhibits metagenesis

154. *Meandrina* belongs to phylum
(1) Platyhelminthes
(2) Ctenophora
(3) Cnidaria
(4) Porifera

155. Which of the following statement is **not correct** for *Taenia*?
(1) Flame cells help in excretion and osmoregulation
(2) They absorb nutrients from host directly through their body surface
(3) Hooks and suckers are absent
(4) Fertilisation is internal and development is through many larval stages

156. Bioluminescence is the characteristic of members of phylum
(1) Ctenophora (2) Mollusca
(3) Annelida (4) Chordata

157. Excretion takes place through Malpighian tubules in the members of phylum which includes
(1) King crab (2) Earthworm
(3) Hookworm (4) Chiton

158. Which of the following possess closed circulatory system?
(1) *Cucumaria*
(2) *Chaetopleura*
(3) *Aplysia*
(4) *Petromyzon*

159. Which is common among *Catla* and *Naja*?
(1) Viviparity
(2) Homoiothermy
(3) Internal fertilisation
(4) Scales on their body

160. A are marine but migrate for B to fresh water. Choose the option that **correctly** fill the blanks A and B.
(1) Urochordates, metamorphosis
(2) Cyclostomes, spawning
(3) Coelenterates, metagenesis
(4) Arthropods, moulting

161. Compared to *Amphioxus* the *Pheretima*
(1) Possess notochord during embryonic development
(2) Possess heart in dorsal side of the body
(3) Possess true coelom and segmentation
(4) Possess closed circulatory system

162. Select the organism where its mouth contains a file-like rasping organ for feeding?
(1) *Nereis*
(2) *Saccoglossus*
(3) *Ophiura*
(4) *Sepia*

163. Metamerism, enterocoelom and close circulatory system are characteristic of
(1) *Ancyclostoma*
(2) *Culex*
(3) *Asterias*
(4) *Bungarus*

164. Choose the **mismatch**.

(1)	Protonephridia	–	Tunicata
(2)	Probocis gland	–	Hemichordates
(3)	Visceral hump	–	Molluscs
(4)	Corals	–	Coelenterata

165. Read the following statements and choose the **correct** answer.
A. The members of Chondrichthyes are mostly predaceous and possess powerful jaws.
B. True fishes possess paired fins.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

166. Match the column-I with column-II and mark the **correct** answer.

	Column-I		**Column-II**
a.	*Pteropus*	(i)	Possess beak
b.	*Struthio*	(ii)	6-15 pairs of gill slits
c.	*Myxine*	(iii)	Pinnae are present
d.	*Antedon*	(iv)	Water vascular system

(1) a(iv), b(iii), c(i), d(ii)
(2) a(iv), b(ii), c(iii), d(i)
(3) a(iii), b(i), c(ii), d(iv)
(4) a(iii), b(ii), c(iv), d(i)

167. Identify the statements as true **(T)** or false **(F)** and select the **correct** option.
(i) Poriferans possess characteristic flagellated choanocytes.
(ii) Aschelminthes are pseudocoelomate and included parasitic as well as non-parasitic roundworms.
(iii) All chordates possess notochord either throughout their life or during early embryonic life.
(iv) Most of the vertebrates do not possess jaws in their mouth.

	(i)	(ii)	(iii)	(iv)
(1)	T	T	F	F
(2)	T	T	T	F
(3)	T	F	T	F
(4)	T	F	T	T

168. Which one is **not** related with coelenterates?
(1) Gastrovascular cavity
(2) Mesolea
(3) Chidoblast
(4) Comb plates

169. Choose the **odd** one w.r.t. bisexual animals.
(1) *Ctenoplana* (2) *Spongilla*
(3) *Fasciola* (4) *Wuchereria*

170. Identify the animal in the given diagram and select the **incorrect** description about it



(1) It possess two pairs of limbs
(2) It's skin is moist and suitable for cutaneous respiration
(3) It can live in aquatic as well as terrestrial habitat
(4) It's alimentary canal opens to exterior by anus

171. Bony fishes possess all of the following features **except**
(1) Terminal mouth
(2) Scales
(3) Three chambered heart
(4) Kidneys for osmoregulation and excretion

172. Which of the following is **incorrect** for sea cucumber?
(1) Belongs to phylum Echinodermata
(2) Triploblastic and coelomate
(3) Possess water canal system
(4) Undergoes external fertilisation

173. Read the following statements and select the **correct** option.

Statement (A) : Sponges exhibit cellular level of organisation and some division of labour occur among the cells.

Statement (B) : When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry

(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

174. Members of which of the following phylums do **not** exhibit similar symmetry in their adult stage?
(1) Annelida, Arthropoda
(2) Arthropoda, Echinodermata
(3) Hemichordata, Chordata
(4) Mollusca, Aschelminthes

175. In which of the following phylums, most of the members are asymmetrical?
(1) Coelenterata
(2) Ctenophora
(3) Platyhelminthes
(4) Porifera

176. Select the organism which is non-poisonous to humans?
(1) *Vipera* (2) Krait
(3) *Hemidactylus* (4) *Trygon*

177. Digestion is both extracellular and intracellular, in all of following **except**
(1) *Ctenoplana* (2) *Gorgonia*
(3) *Adamsia* (4) Scypha

178. Which of the following phylum has organisms with a soft body surrounded by an external calcareous shell?
(1) Molluscs (2) Arthropods
(3) Echinoderms (4) Hemichordates

179. Heart pumps mixed blood in systemic aorta in
(1) Reptiles (2) Mammals
(3) Aves (4) Fishes

180. What is **not** true about *Balanoglossus, Pinctada* and *Laccifer*?
(1) All possess dorsal, hollow and single central nervous system
(2) All possess bilateral symmetry
(3) All possess true coelom
(4) All possess organ system level of organisation

❑ ❑ ❑

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